

MAE SARIANG, Thailand

WMO: 483250

Lat: 18.175N

Lon: 97.934E

Elev: 700

StdP: 14.33

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	53.2	55.8	49.2	53.2	66.6	51.8	58.6	67.5	7.1	85.8	5.6	85.1	0.0	N/A	0.277

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	24.8	101.4	76.0	99.2	75.3	97.1	74.9	80.8	90.7	80.1	89.7	79.6	88.9	1.7	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.3	151.1	84.4	77.9	148.6	83.8	77.3	145.8	83.2	45.1	91.0	44.4	89.9	43.8	89.4	87.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.7	6.3	3.8	DB	48.9	104.8	3.3	2.2	46.5	106.4	44.6	107.7	42.8	109.0	40.4	110.6	(4)
(5)			WB	47.7	83.5	3.1	1.8	45.5	84.8	43.6	85.8	41.9	86.8	39.6	88.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	80.3	73.8	75.6	81.0	86.5	85.3	82.1	80.7	80.6	81.9	82.0	79.2	74.9	(6)
(7)		DBStd	4.71	3.04	3.29	3.16	3.06	3.71	2.49	2.11	1.96	1.95	1.99	3.20	3.81	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	2	0	0	0	0	0	0	0	0	0	0	0	2	(9)
(10)		CDD50	11066	738	718	960	1095	1093	962	953	950	958	991	877	771	(10)
(11)		CDD65	5593	273	298	495	645	628	512	488	485	508	526	427	308	(11)
(12)		CDH74	53071	1983	3015	5712	8210	7248	4837	4139	3988	4478	4361	3103	1997	(12)
(13)		CDH80	23095	790	1563	3350	4841	3577	1592	1118	1027	1480	1695	1313	749	(13)
(14)	Wind	WSAvg	0.3	0.1	0.3	0.5	0.6	0.5	0.5	0.4	0.3	0.2	0.2	0.2	0.2	(14)
(15)	Precipitation	PrecAvg	50.5	0.4	0.1	0.2	1.3	7.5	7.0	8.6	10.3	8.2	5.6	0.8	0.6	(15)
(16)		PrecMax	63.2	3.8	0.9	2.7	3.7	12.6	13.0	14.3	16.5	14.9	11.3	3.4	3.3	(16)
(17)		PrecMin	42.5	0.0	0.0	0.0	0.0	1.1	3.7	5.5	5.6	3.0	1.4	0.0	0.0	(17)
(18)		PrecStd	5.6	1.0	0.2	0.7	1.3	2.9	2.3	2.4	2.7	3.1	3.1	0.9	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.2	97.3	100.8	104.4	103.8	95.1	92.8	91.6	93.5	94.2	93.5	91.5	(19)
(20)			MCWB	71.7	71.4	72.8	76.1	77.7	79.4	79.4	79.4	79.8	79.0	77.4	74.8	(20)
(21)		2%	DB	89.8	94.8	98.9	102.2	100.1	92.2	89.9	89.2	91.4	92.3	91.6	89.1	(21)
(22)			MCWB	71.1	71.0	72.4	76.0	78.1	79.2	78.7	78.9	79.5	79.0	76.8	73.6	(22)
(23)		5%	DB	87.6	92.5	97.2	100.4	97.1	89.8	87.7	87.3	89.6	90.8	89.9	87.2	(23)
(24)			MCWB	70.6	70.5	72.3	75.9	78.0	78.7	78.2	78.4	79.1	78.7	76.2	72.8	(24)
(25)		10%	DB	85.0	89.9	95.2	98.3	94.1	87.5	85.6	85.4	87.6	88.9	87.9	84.8	(25)
(26)			MCWB	69.7	69.8	72.0	75.8	78.1	78.3	77.8	78.0	78.6	78.2	75.4	71.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.3	76.3	77.8	80.7	81.4	81.0	80.6	80.7	81.2	81.0	80.0	77.3	(27)
(28)			MCD B	86.9	91.2	94.6	98.7	93.3	89.8	89.1	88.3	90.0	90.5	90.1	87.6	(28)
(29)		2%	WB	73.7	73.9	76.2	79.1	80.2	80.0	79.5	79.6	80.3	80.1	78.6	75.6	(29)
(30)			MCD B	84.8	89.2	92.6	96.0	92.3	88.5	87.1	86.7	88.6	89.4	88.2	85.9	(30)
(31)		5%	WB	72.4	72.4	75.0	78.2	79.5	79.3	78.8	78.9	79.6	79.4	77.5	74.2	(31)
(32)			MCD B	83.1	87.4	90.6	94.1	91.4	87.0	85.6	85.5	87.5	88.4	86.7	83.9	(32)
(33)		10%	WB	71.0	71.0	73.8	77.4	78.8	78.7	78.1	78.2	78.9	78.7	76.4	72.9	(33)
(34)			MCD B	81.5	85.6	88.8	92.3	90.5	85.8	84.2	84.1	86.3	87.2	84.8	81.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	28.7	34.1	31.3	24.8	17.0	11.8	10.5	10.5	12.9	16.2	20.6	24.3	(35)
(36)			MCD BR	30.4	35.8	33.3	27.5	22.0	15.2	13.6	13.4	15.6	18.3	21.6	25.0	(36)
(37)			MCWBR	12.7	13.4	10.2	6.8	5.1	4.7	4.6	4.8	5.5	6.6	8.0	10.2	(37)
(38)		5% WB	MCD BR	26.5	31.7	28.1	23.8	17.9	13.0	11.9	11.8	14.4	16.5	19.1	22.3	(38)
(39)			MCWBR	11.9	12.7	9.3	6.7	5.5	4.7	4.6	4.8	5.5	6.6	7.6	9.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.419	0.480	0.639	0.655	0.523	0.456	0.442	0.449	0.469	0.511	0.437	0.420	(40)	
(41)		taud	2.259	2.099	1.708	1.742	2.121	2.348	2.390	2.358	2.284	2.120	2.287	2.284	(41)	
(42)		Ebn at Noon	272	261	225	222	251	266	270	270	265	249	265	268	(42)	
(43)		Edh at Noon	41	50	77	75	51	40	38	40	43	49	40	39	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1518	1774	1872	1932	1634	1312	1202	1209	1378	1463	1509	1434	(44)	
(45)		RadStd	69	53	116	125	148	115	139	88	65	81	99	77	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.58	N/A	N/A	N/A	+0.41	+0.67	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		+0.49	N/A	N/A	N/A	+0.22	+0.34	N/A	N/A	+265	+266

Nomenclature: See separate page